

The Unified Nucleosynthesis Architecture: Simulating Topological Fusion and Nuclear Benevolence

Version 2.0 — Revised & Scientifically Deepened | June 2026

Classification: Tardigradia Game Engine | Physics Deep Dive Series | TGPU v2.0 **Particle Class:** Nuclear fusion | pp chain | CNO cycle | r-process | s-process | Neutron star merger nucleosynthesis
Source: Extends v1.0Particle Fusion Deep Dive Request.pdf (V1.0)

Δ Change Log V1.0 → V2.0

Parameter	V1.0	V2.0	Source
ITER milestone	Fusion progress noted	V2.0: NIF (National Ignition Facility) 2023: ignition achieved and confirmed. 3.15 MJ output from 2.05 MJ laser input. Q>1 reached. Fusion energy now validated. V2.0: FUSION_IGNITION event type added — triggers when local proton density exceeds ignition threshold and temperature > 10 keV	NIF Science 2023; ITER 2024 progress
r-process in neutron star merger	General r-process mention	V2.0: Kilonova AT2017gfo (GW170817): gold, platinum, lanthanide abundances measured. r-process confirmed in NS-NS mergers. V2.0: NS_MERGER event creates burst of heavy nuclei (A>130). Strontium-88 and barium-138 yield data from MUSE spectrograph 2023	MUSE 2023 kilonova; Cowan et al. review

Parameter	V1.0	V2.0	Source
pp chain cross-section measurement	Solar pp chain	V2.0: LUNA collaboration (Gran Sasso 2022-2024): measured ${}^3\text{He}({}^3\text{He}, 2p){}^4\text{He}$ at solar energies 20-30 keV. S-factor $S_{33}(0) = 5.12 \text{ MeV}\cdot\text{b}$ (3.6% more precise than V1.0 value). V2.0: S-factor lookup table updated for all pp chain reactions	LUNA Collaboration 2022
Quantum tunneling in stars	Gamow peak described	V2.0: Gamow peak energy for pp fusion: $E_G = (\pi\alpha_{\text{rc}}/2)^{2/3} \times (k_B T)^{1/3}$. At $T_{\text{Sun}} = 15 \times 10^6 \text{ K}$: $E_G = 5\text{-}20 \text{ keV}$ window. In simulation: fusion probability computed as $\exp(-2\pi\eta) \times \text{geometric cross-section at Gamow energy}$ — tunneling essential	Fenzy review 2024

1. The One-Particle Universe Framework — V2.0 Update

The foundational architecture established in V1.0 — Worldline Monism, the “Origami Universe” metaphor, and the Object-Oriented ontology distinguishing intrinsic from extrinsic properties — is **fully preserved** and remains physically rigorous. V2.0 layers NEW discoveries from 2022-2026 atop this framework.

1.1 WebGPU Compute Architecture (V2.0 Core Infrastructure)

V1.0 designed for WebGL/Three.js on the CPU-side particle update loop. V2.0 upgrades to WebGPU (available in Chrome 113+ May 2023, Firefox 2024):

```
// WGSL Compute Shader - V2.0 Particle Integration
@binding(0) @group(0) var<storage, read_write> particles: array<ParticleState>;

struct ParticleState {
    position: vec4f,
    momentum: vec4f,
    spin: vec2f,           // Spinor components
    berry_phase: f32,      // Geometric phase accumulated
    proper_time: f32,      // tau - worldline parameter
    color_charge: u32,      // Packed color in 2 bits (R=0,G=1,B=2)
```

```

    quantum_numbers: u32, // Packed: isospin, strangeness, charm, beauty
    decay_lifetime: f32,
    entanglement_id: i32, // -1 = unentangled; >=0 = entanglement group
}

@compute @workgroup_size(64)
fn updateParticles(@builtin(global_invocation_id) id: vec3u) {
    let i = id.x;
    if (i >= arrayLength(&particles)) { return; }

    var p = particles[i];

    // Feynman-Schwinger proper time evolution
    let dtau = uniforms.dt / p.mass; // proper time step

    // Update worldline parameter
    p.proper_time += dtau;

    // Berry phase accumulation (topological phase)
    p.berry_phase += dot(uniforms.gauge_field, p.momentum.xyz) * dtau;

    // Lorentz force (for charged particles)
    let F = cross(uniforms.B_field, p.momentum.xyz) * p.charge_sign;
    p.momentum.xyz += F * uniforms.dt;
    p.position.xyz += p.momentum.xyz * uniforms.dt / p.energy;

    particles[i] = p;
}

```

1.2 Updated Physical Constants (CODATA 2022)

Constant	V1.0 Value	V2.0 Value (CODATA 2022)
m _e	9.10938356×10 ⁻³¹ kg	9.1093837139×10 ⁻³¹ kg
e	1.60217663×10 ⁻¹⁹ C	1.602176634×10 ⁻¹⁹ C (exact)
ħ	1.054571817×10 ⁻³⁴ J·s	1.054571817×10 ⁻³⁴ J·s (exact)
c	2.99792458×10 ⁸ m/s	2.99792458×10 ⁸ m/s (exact)
α	1/137.036	1/137.035999084 ±0.000000021

2. V2.0 Enhanced Data Structures

Extending V1.0 struct definitions with 2022-2026 discoveries:

```

// V2.0 Universal Particle State – extends V1.0
struct ParticleInstanceV2 {
    // Inherited from V1.0
    double proper_time; // tau worldline parameter (meter)
    glm::dvec3 position; // 3-space vector
    glm::dvec3 momentum; // 3-momentum (relativistic)
    float spin_z; // +0.5 or -0.5 (or eigenvalue)
    float berry_phase; // Accumulated Berry/Aharonov-Bohm phase
}

```

